# 1. Introduction

Single-Molecule Localization Microscopy (SMLM) enables nanometer-scale fluorescence imaging. As output, it delivers datasets that are naturally represented as localization coordinate lists or point clouds and their localization uncertainties. Transforming these coordinate lists into biological insight requires quantitative model fitting because the relevant structural parameters are not directly observed but must be inferred from the spatial distribution of localizations A central difficulty is that SMLM localizations are affected by multiple sources of uncertainty. Besides localization imprecision there are additional deviations that cannot be captured by the localization precision alone like linkage effects and experimental instabilities. Consequently, meaningful structure inference depends on a scoring function that incorporates the uncertainty into the structural hypothesis to the observed localizations in a meaningful way. In practice, computational cost becomes a limiting factor, because model fitting or probabilistic sampling approaches require repeated evaluation of the scoring function for many candidate models (Wu et al. 2023).

This thesis addresses this methodological bottleneck by developing and benchmarking likelihood-based scoring restraints for fitting SMLM localization data to a structural reference model. As framework the Integrative Modeling Platform (IMP) is used. IMP provides a general workflow for integrative structural modeling. After the input information is processed and translated into a model representation and a scoring function, good-scoring models are searched and subsequently validated (Hancock et al. 2022). Within this workflow, the present work focuses on the representation-and-scoring component by implementing alternative scoring strategies and assessing their computational efficiency and practical behavior on SMLM-like data. In this work, SMLM localizations are compared to a Nuclear Pore Complex (NPC) structural model. Uncertainties in fluorophore positions are represented using Accessible Volumes to reflect plausible labeling geometries. Several scoring functions are then evaluated, including distance-based approaches and distribution-based approaches, with the aim of finding implementations which remain feasible for large datasets.

## 1.1. Concepts and Resolution

Microscopy has undergone rapid development in the last three decades. Fluorescence-based microscopy in particular has enabled the observation of proteins at increasingly small length scales (Klar et al. 2000). Resolution is defined as the minimum distance at which two objects can still be distinguished as distinct entities (Koho et al. 2019). For a long time, the diffraction limit formulated by Ernst Abbe in 1873 was considered a fundamental barrier for light microscopy (Huang, Babcock, and Zhuang 2010). In a classical far-field microscope, the diffraction limit can be expressed in the lateral (x–y) plane as

$$d_{xy} = \frac{\lambda}{2n\sin(\alpha)} \qquad \text{(eq. 1)}$$

and in the axial (z) direction as

$$d_z = \frac{\lambda}{2n\sin(\alpha)^2} \qquad \text{(eq. 2).}$$

Here, $\lambda$ denotes the wavelength of light, $\alpha$ the semi-aperture angle of the objective lens and $n$ the refractive index of the imaging medium (Ehrenberg 2014). Approximately, the attainable resolution is on the order of half the wavelength of the light. The expressions above highlight the dependence on refractive index and aperture angle which is a property of the microscope. In conventional fluorescence microscopy, this implies that objects separated by roughly 200–300 nm can typically not be distinguished laterally, while the required separation along the optical axis is usually larger (Huang, Babcock, and Zhuang 2010).

As a direct consequence of the diffraction limit a point-like emitter is not observed as a mathematical point but as a blurred intensity distribution in the image. This blurring can be described by the convolution of the object with the point-spread function (PSF). A point focused through a microscope is therefore observed as a focal spot with maximal intensity at its center and decreasing intensity with increasing distance. This PSF is often approximated by a Gaussian function (Lelek et al. 2021; Hell 2003; Huang, Babcock, and Zhuang 2010; Thompson, Lew, and Moerner 2012). Consequently, the PSFs of two objects must not strongly overlap for them to be distinguishable, and the relevant separation is commonly related to the full width at half maximum (FWHM) of the PSF, where a smaller FWHM corresponds to better resolution (see FIG. 1) (Thompson, Lew, and Moerner 2012; Lelek et al. 2021).

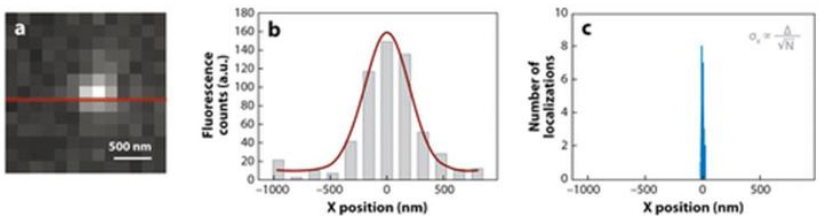



FIG. 1: Thompson 2012

Kommentiert [DR1]: TODO: erstellen

The key conceptual step toward overcoming the diffraction barrier in fluorescence microscopy was enabled by fluorophore photophysics in the 1990s. In particular, the ability to switch fluorophores between emissive and non-emissive states (Moerner and Kador 1989; Dickson et al. 1997). Instead of attempting to resolve spatially overlapping emitters in a single exposure, fluorescence microscopy uses sparse or stochastic emission such that nearby fluorophores are not simultaneously active and their PSFs remain separable. While the width of the PSF remains diffraction-limited, the center position of an isolated PSF can be estimated with a localization precision that is substantially higher than its width. By repeatedly localizing many emitting single fluorophores, a high-resolution representation of the underlying structure can be reconstructed, thereby circumventing the classical Abbe limit (Betzig et al. 2006).

## 1.2. Single-Molecule-Localization-Microscopy

Single-Molecule Localization Microscopy (SMLM) summarizes a group of super-resolution techniques such as Stochastic Optical Reconstruction Microscopy (STORM) and Photoactivated Localization Microscopy (PALM) (Betzig et al. 2006; Rust, Bates, and Zhuang 2006). These methods share the principle of stochastic activation of fluorophores. This means, in each recorded frame, only a sparse subset emits fluorescence. This sparsity ensures that nearby fluorophores are not simultaneously active, so their point-spread functions (PSFs) do not substantially overlap and can be seen as separate emitting signals. The complete super-resolved image is then reconstructed from thousands of individual frames by estimating the emitter positions in each frame and combining these localizations into a global representation. The position of an emitter is determined by fitting a PSF model to the measured intensity distribution of a single fluorophore. This delivers the localization coordinates for each detected event. SMLM activates individual molecules instead of observing ensembles. This provides access to nanoscale spatial information in the form of point clouds (Huang, Babcock, and Zhuang 2010; Liu, Hoess, and Ries 2022).

However, SMLM must balance a trade-off with labeling and activation. On one hand, it must be sufficiently sparse to avoid overlapping PSFs, on the other hand, the labeling density must be sufficient to sample the underlying structure without aliasing. This refers to the Nyquist–Shannon sampling argument that the mean labeling distance should not exceed $\frac{\Delta_{min}}{2}$ with $\Delta_{min}$ denoting the resolution limit defined before (Thompson, Lew, and Moerner 2012). The practical solution to this tension is reversible switching of fluorophores between “on” and “off” states (photoswitching), which enables dense labeling while maintaining sparse emission per frame. In PALM, for example, photoswitching can be induced by UV light irradiation that controls the activation state of photoactivatable fluorophores. By controlling fluorophore states, the data acquisition proceeds iteratively until a high-resolution reconstruction can be assembled from the accumulated localizations (Kikuchi et al. 2023; Huang, Babcock, and Zhuang 2010).

The photophysical basis of these switching processes can be illustrated using a Jablonski diagram, which summarizes transitions between electronic states and includes long-lived dark states that are exploited to temporally separate emission events (FIG. 2).

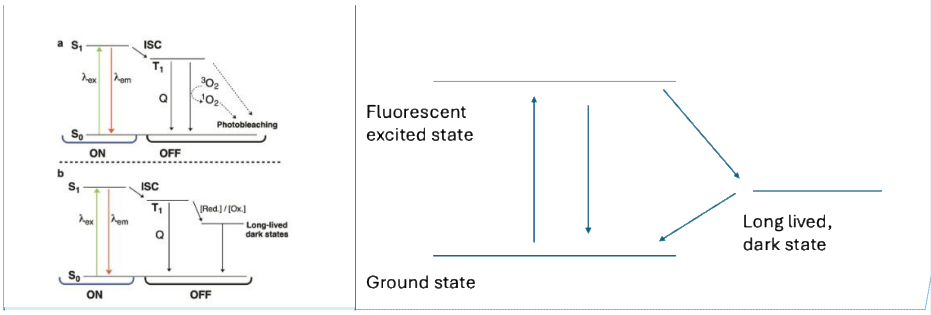



FIG. 2: On state, off state (different types of off states: triplet, radical, photo switching) Adapted from Kikuchi et al. (2023).

**Kommentiert [TP2]:** Eigenes Diagram malen

Due to these comparatively accessible requirements for overcoming the diffraction barrier in fluorescence imaging, SMLM has become a widely used approach for generating super-resolved images in cell biology (Huang, Babcock, and Zhuang 2010). SMLM typically achieves resolutions on the order of 20–30 nm under suitable experimental conditions (Kuang et al. 2021). The resulting data are well suited for quantitative analysis because they consist of localization coordinates and additional localization information such as photon intensity and uncertainty estimates (Weidner et al. 2023; Wu et al. 2023).

## 1.3. Uncertainties in SMLM

Despite the substantial resolution improvements achieved by SMLM, estimated fluorophore positions remain subject to uncertainty. The quality of localization algorithms is typically quantified using the mean squared error (MSE) between the estimated position $\hat{\mathbf{x}}$ and the true position $\mathbf{x}$:

$$\mathrm{MSE}(\hat{\mathbf{x}}) = \mathbb{E}[\| \hat{\mathbf{x}} - \mathbf{x} \|^2] = \mathrm{Var}(\hat{\mathbf{x}}) + \| \mathrm{Bias}(\hat{\mathbf{x}}) \|^2 \quad \text{(eq. 3)}$$

The decomposition of the MSE explicitly separates the unsystematic statistical error (variance) from the systematic error (bias).

The statistical error is described by the variance and reflects the precision of the localization that is the spread of repeated measurements around their mean due to photon counting noise (Lelek et al. 2021).

$$Var(\hat{x}) = \langle (\hat{x} - \langle \hat{x} \rangle)^2 \rangle = E\left[\left(\hat{x} - E\left(\left[\hat{x}\right]\right)\right)^2\right] \qquad \text{(eq. 4)}$$

In SMLM, this localization precision is primarily dependent on the number of detected photons $N$ and the diffraction-limited width of the point-spread function (PSF). It is often approximated as (Kikuchi et al. 2023; Lelek et al. 2021; Ehrenberg 2014):

$$\sigma_{\mathrm{loc}} \propto \frac{d_{\mathrm{Abbe}}}{\sqrt{N}} \quad \text{(eq. 4)}$$

This proportionality relation describes a theoretical lower bound for the precision named the Cramér–Rao lower bound (CRLB). It represents the best achievable precision for an unbiased estimator under the assumption of an ideal noise model (Deschout et al. 2014).

Systematic errors define the bias and therefore affect the accuracy of the estimated positions. It quantifies the systematic deviation of an estimator $\hat{x}$ from the true parameter $x$ (Lelek et al. 2021):

$$B(\hat{x}) = x - \langle \hat{x} \rangle = x - E(\hat{x}) \text{ (eq. 5)}$$

Typical sources include sample drift because of thermal or mechanical effects of the microscope and fluorescence-related effects like linkage error. Linkage error arises because the fluorophore

is physically displaced from the target molecule by the labeling complex. For binding these fluorophores to the molecule often antibodies are used. This displacement is not a fixed vector but rather a probability distribution over the volume accessible to the dye. For immunolabeling, the mean linkage error can be substantial: approximately 13 nm for primary antibodies and up to 20 nm when using secondary antibodies (FIG. 3) (Liu, Hoess, and Ries 2022; Lelek et al. 2021).

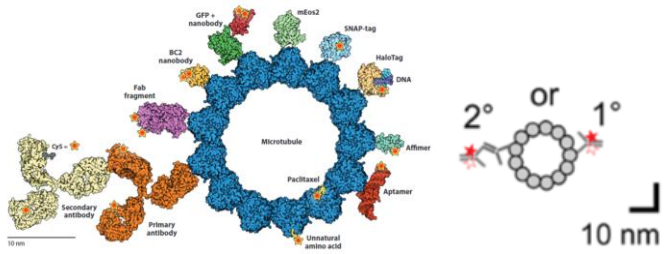
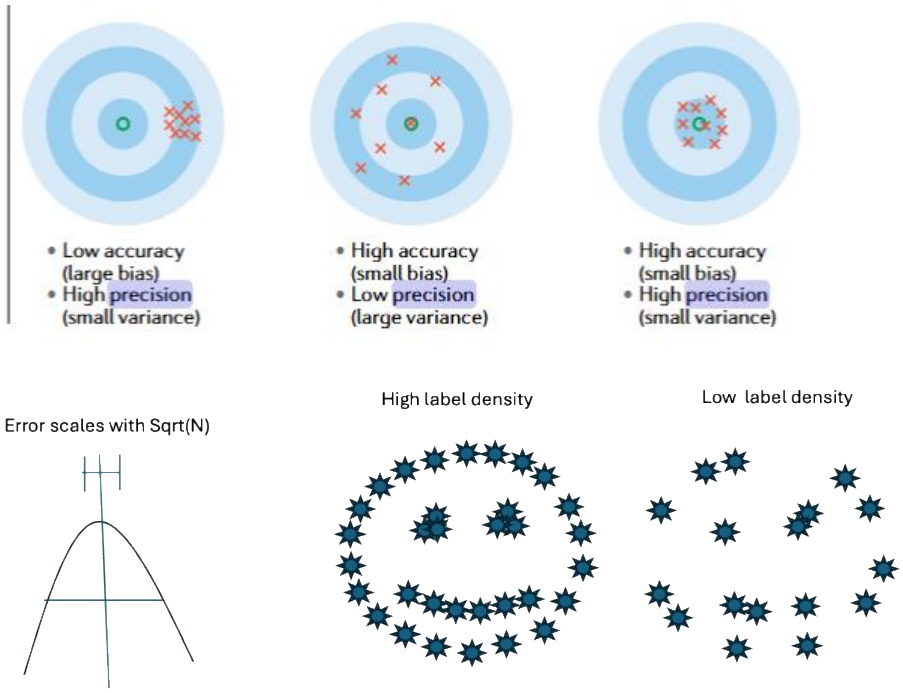



FIG. 3: Liu et al. 2022

The connection between the different types of uncertainty is shown in figure 4.



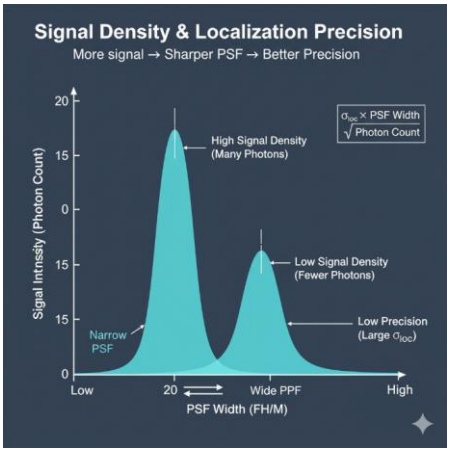



FIG. 4: TODO

Systematic errors can be mitigated experimentally (e.g., via drift correction or smaller labels like nanobodies) or handled computationally by incorporating explicit error models into the analysis. While high photon counts can minimize statistical error, accurate structural modeling requires a rigorous treatment of these systematic biases (Liu, Hoess, and Ries 2022; Lelek et al. 2021).

## 1.4. Modeling

**Kommentiert [DR3]:** Für alles theoretische nochmal Raveh schauen. Sehr, sehr gut für Theorie.

High-resolution microscopy has evolved from a descriptive and qualitative science to a quantitative one, as data from SMLM experiments are well-suited for mathematical processing (Hugelier, Colosi, and Lakadamyali 2023; Wu et al. 2023). However, fluorescence-based techniques remain often inherently biased. As described in Chapter 1.3, structures of interest are never observed directly, but rather are inferred indirectly by detecting fluorophores, which introduces systematic effects such as linkage error. While such biases can be minimized experimentally, they can also be corrected computationally by explicitly describing the experimental process within the model.

A model in this case is a mathematical representation of a biological system used to explain existing information and predict the outcomes of future experiments (Kinz-Thompson, Ray, and Gonzalez 2021; Rout and Sali 2019). For SMLM, it is crucial to distinguish between the acquired experimental localization data and the structural model, which represents the underlying molecular arrangement explaining that data.

Often, the available data is insufficient to determine a unique structure. Consequently, the goal is rarely to find a single model, but rather an ensemble of models that are all consistent with the given information. To evaluate this consistency, a scoring function is required that quantifies the agreement between a model and the experimental observations (Joseph et al. 2017; Koukos and Bonvin 2020).

A robust scoring function typically consists of two components. The forward model, which predicts the data that would be observed for a specific structure in the absence of noise, and the noise model, which quantifies the deviation between this prediction and the actual experimental measurements (Rieping, Habeck, and Nilges 2005; Peulen and Sali 2023). Nuisance parameters can be integrated into the model as well. These are variables that are not of direct interest but are necessary to model experimental uncertainties like background noise or specific labeling inaccuracies (Rout and Sali 2019). Additionally restraints can be incorporated into the scoring function, for example, in the form of penalty terms that penalize violations of known geometries and reduce the sample space (Joseph et al. 2017).

The scoring function can be utilized within distinct statistical frameworks. In a frequentist approach, parameters are treated as fixed but unknown constants. The goal is typically to identify the single set of parameters that maximize the probability of the observed data. Probability in this case can be interpreted as a long-run frequency of the outcome. A central concept is the likelihood $\mathcal{L}(\theta|D) \equiv p(D|X)$ which quantifies the frequentist probability for data $D$ under the assumption of parameter $\theta$/model $X$. A widely used method to find the parameter set that maximizes this likelihood is Maximum Likelihood Estimation (MLE). MLE as a point estimator estimates a single best result (Kinz-Thompson, Ray, and Gonzalez 2021; Cox 2006). While effective for fitting tasks where a single best solution exists, MLE can be limiting for complex, underdetermined biological systems where multiple different structures fit the data equally well. The frequentist interpretation of the likelihood as the probability of data given parameters might be seen as contra-intuitive because the actual problem that is to be solved is to find the probability for parameters or models for given data (Kinz-Thompson, Ray, and Gonzalez 2021). For analytical reasons the log-likelihood is often used instead of the normal likelihood.

Bayesian inference offers a rigorous alternative. Finding the right model is an inverse problem which means that the determination of a structure from data is impossible because of random influences. The use of probabilities in a Bayesian approach can handle these uncertainties by ranking models with probabilities instead of looking for one particular model that has the highest likelihood (Rieping, Habeck, and Nilges 2005). In the Bayesian view, parameters are treated as random variables that have an associated probability distribution (MacKenzie et al. 2018; Rout and Sali 2019). Instead of searching for a single "best" structure, Bayesian modeling ranks all possible models based on their posterior probability. According to Bayes' theorem, the posterior probability is defined as:

$$p(X, \xi|\mathrm{D}, \mathrm{I}) \propto \mathrm{p}(\mathrm{D}|\mathrm{X}, \xi, \mathrm{I})\mathrm{p}(\mathrm{X}|\mathrm{I})\mathrm{p}(\xi|\mathrm{I}) \qquad \text{(eq. 6)}$$

with $\xi$ as non-observable independent nuisance parameter. In contrast to the frequentist approach, here, the posterior $p(X, \xi|\mathrm{D}, \mathrm{I})$ is sought which is the probability of a model $X$ given prior knowledge I and experimental data D. This probability is exactly the scientific goal of the modeling process as discussed regarding the frequentist likelihood view. The posterior is proportional to the product of previously existing knowledge, the prior $p(X|\mathrm{I})$, and the newly obtained information from the experiment, the likelihood $p(D|\mathrm{X}, \xi, \mathrm{I})$ which includes the forward and noise model. The probability of the parameter, which is known from the prior, can

Kommentiert [DR4]: X gegen \theta tauschen.

be adjusted even with small amounts of data obtained in an experiment using the new findings from the likelihood. The posterior is thus a kind of metric of how the experiment updates the prior (Rieping, Habeck, and Nilges 2005; Kinz-Thompson, Ray, and Gonzalez 2021; Rout and Sali 2019). The Bayesian scoring function itself is defined as the negative logarithm of the posterior density (Bonomi et al. 2019):

Kommentiert [DR5]: Immer so definiert? Prüfen.

Kommentiert [DR6R5]: Noch nuisance rein

$$S(X,\xi) = -\log p(X,\xi|D,I) = -\log[p(D|X,\xi,I)p(X|I)] \quad \text{(eq. 7)}$$

If a solution cannot be found analytically, which is often the case for complex biomolecular systems, stochastic approximation approaches like Markov Chain Monte Carlo sampling (MCMC) must be used for generating the ensemble of good-scoring models. Finally, this ensemble is validated against data that was not used to develop the model. For example, validations with different datasets or against a known ground truth can be used to check whether the model also provides valid results with other inputs. (Russel et al. 2012; Saltzberg et al. 2021; Habeck 2017; Kinz-Thompson, Ray, and Gonzalez 2021). Figure 5 illustrates the two different approaches.

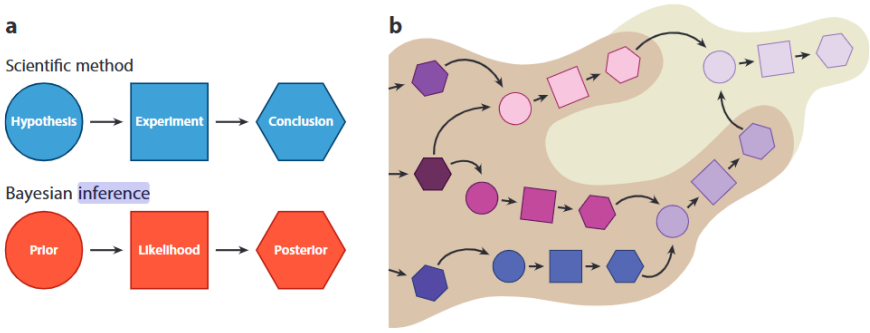



FIG. 5: Comparison (Kinz-Thompson)

These sampling procedures that are  can become a bottleneck in Bayesian inference (Habeck 2017). Therefore, the improvement of efficiency is crucial. One possible option could be divide-and-conquer approaches to cut bigger problems into smaller parts that are easier to solve. Especially when considering the possible amount of data and its dimensionality it gets clear how important efficient scoring is because of often computational expensive operations when calculating matrices in distributions (Bonomi et al. 2019; Latham et al. 2025; Peulen and Sali 2023).

Finally, Bayesian modeling provides a framework that can be used iteratively. The explicit use of prior and likelihood makes it easy to connect known data and new experimental data. The posterior becomes the prior in subsequent experiments and so on. Additionally, this also improves the possibility to incorporate knowledge from various sources and techniques into a model on a scientific basis. The use of probabilities offers an approach to implement uncertainty

Kommentiert [DR7]: Uncertainty muss auf jeden Fall rein für Bayesian vs. frequentist mit Bezug auf Kap. Unsicherheit

into the model. This could be helpful for handling incomplete and uncertain data as in SMLM experiments. Instead of giving just a single solution as achieved with point estimators, a set of ranked possible solutions are gained by using a Bayesian approach. As new information can be incorporated into the model, the model becomes more accurate (Kinz-Thompson, Ray, and Gonzalez 2021; Rieping, Habeck, and Nilges 2005).

# 2. Materials and Methods

Kommentiert [DR8]: Hier noch Workflow

Kommentiert [DR9]: Achtung: wie bekomme ich den turn zu drei am besten hin frequentist vs.bayesian

Kommentiert [DR10R9]: Wenn ich validation in die diskussion packe. Habe ich imp als reinen container, der nur ein paar restraints enthält. Kein posterior sondern nur mle durch distance.

As framework for the modeling process the Integrative Modeling Platform (IMP), an open-source Python package designed for the structural characterization of complex biological systems, was chosen. It was developed to make integration of diverse data sources into modeling easier by offering a standardized framework that can be adapted to the specific data. IMP uses a modular structure which facilitates the integration of different scoring functions and model representations, allowing for a flexible modeling pipeline tailored to the specific question (Russel et al. 2012; Sali 2021).

Kommentiert [DR11]: Gut! Hier ist nichts über IMP und Bayesian drin. Das ist gut hinsichtlich der Result: Wie leite ich das ambesten ein - frequent vs. bayesian. Wie entwicklung der scoring?

## 2.1. Gathering Information

The first step in modeling is the decision about the data used. This initial stage involves both gathering and processing the primary data sources. Two data sources are necessary for the scoring function.

First, the experimental data. Since this data could not be generated experimentally for this work, the super-resolution microscopy data was obtained from the ShareLoc platform, a public repository for single-molecule localization microscopy (SMLM) datasets that adheres to FAIR principles (Findable, Accessible, Interoperable, and Reusable) (Ouyang et al. 2022). The chosen dataset contains localizations of the gp210 protein in the *Xenopus laevis* nuclear pore complex (NPC), stained with the fluorophore Alexa 647 (Löschberger et al. 2012). The data, provided in CSV format, includes the x and y coordinates (in nanometers), frame number, amplitude for each detected fluorophore and fit residues. The amplitude, representing fluorophore intensity, and the fit residues serve as indicators of localization precision. Using the discussed assumptions regarding the relation of photon count and precision, the precision can be added to each data point by applying (Wu et al. 2023):

$$\sigma_{loc} \geq \frac{\sigma_0}{\sqrt{N}} \Rightarrow \sigma \approx precision \approx \frac{1}{\sqrt{Amplitude}} \quad \text{(eq. 8)}$$

Second, the structural model data that serve as a ground truth. A structural model of the Nuclear Pore Complex was sourced from the Protein Data Bank (PDB). The PDB is an archive for 3D structural data of macromolecules which is committed to the FAIR principles, too. The specific structure used was that of the yeast NPC (PDB ID: 7N85). This model was determined using cryo-electron tomography and integrative modeling techniques (Akey et al. 2022).

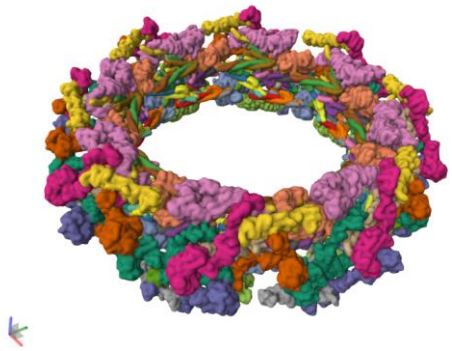


Figure 8: TODO: NPC PDB

The structural data were provided in the macromolecular Crystallographic Information File (mmCIF) format, which contains atomic coordinates in an Ångström-based Cartesian system. Since the NPC structural data managed on PDB is reviewed, it can be taken as a reference for the NPC. Due to the complexity of this format, standard Python libraries are insufficient for direct processing. Therefore, the Integrative Modeling Platform (IMP) was used to parse and handle these data, enabling the creation of model representations suitable for comparison with the SMLM data (wwPDB consortium 2019).

## 2.2. System Representation and Scoring Function

Kommentiert [DR12]: Genauere Ausarbeitung der Scoring Funktion im Results Kapitel.

IMP (Russel et al. 2012) was selected as the integrative modeling framework because it provides a modular architecture for constructing model representations and assembling scoring restraints into a composite scoring function. The IMP workflow decouples data representation from scoring, allowing different scoring strategies to act on a shared structural representation.

The structural data was processed in IMP. It was parsed from the mmCIF file and loaded into IMP using standard parsing routines. To account for the spatial displacement of fluorophores from the protein backbone due to the labeling complex (antibodies and linker), fluorophore attachment sites were represented using Accessible Volumes (AVs). An AV represents the spatially extended region where a fluorophore can be positioned relative to a fixed attachment point on the protein, subject to geometric constraints (linker length, steric exclusion). After AV generation, each AV was resampled to produce a discrete set of representative coordinates for use in subsequent scoring calculations. These coordinates serve as the "forward model" representation of fluorophore positions: they encode plausible dye positions given the labeling geometry and the structural model. The code in BOX 1 shows the creation of accessible volumes at the NPC. First, we decide which position in the hierarchy of the NPC the AV is to be attached to. Then the parameters that describe the AV must be defined so that the AV is a spatial representation of all possible positions a fluorophore can occupy relative to its attachment point on the protein. The resolution can be defined to fit the scale of the data here

as well. IMP offers methods for handling structural data (IMP.Atom), for managing model components (IMP.Core) and generating Accessible Volumes (IMP.bff).

```
m = IMP.model(NPC.cif)
hier = IMP.atom.read_mmcif(NPC.cif, m)
chains = [0, 1, 2, 3, 4, 5, 6, 7]
avs = list()
for chain in chains:
    av_particle = IMP.Particle(m)
    sel = IMP.atom.Selection(hier)
    sel.set_chain_id(chain)
    sel.set_atom_type(IMP.atom.AtomType("CA"))
    sel.set_residue_index(133)
    source = sel.get_selected_particles()[0]
    av_parameter = {
    "linker_length": 150.0,
    "radii": (30.0, 0.0, 0.0),
    "linker_width": 4.0,
    "allowed_sphere_radius": 10.0,
    "contact_volume_thickness": 0.0,
    "contact_volume_trapped_fraction": -1,
    "simulation_grid_resolution": 8.0
    }
    IMP.bff.AV.do_setup_particle(m, av_p, source, **av_parameter)
    av = IMP.bff.AV(m, av_p)
    av.resample()
    avs.append(av)
    mean = IMP.core.XYZ(av)
```

BOX 1: TODO – keep it? – and more precise – Parameter angeben - Linker length 150.0 Å | Linker width 4.0 Å | Dye radius 30.0 Å | Allowed sphere radius 10.0 Å | Simulation grid resolution 8.0 Å | Contact volume thickness 0.0 Å | Contact volume trapped fraction −1 —

Experimental data from ShareLoc is expressed in nanometers, while IMP internally uses Ångströms. A unit conversion was applied consistently throughout the pipeline before any distance calculations.

## 2.3. Scoring function

Three alternative scoring functions were implemented in IMP as restraints. Each function takes as input the resampled model AV coordinates, the experimental localization coordinates, the uncertainty derived from the amplitude and parameters for alignment and scaling.

First, a pointwise distance-based scoring that evaluates the spatial distance between experimental localizations and model AV positions. This approach treats each localization as an independent observation.

Second, Neighborhood-based scoring. Rather than comparing each model point to all data points, only within a specified search radius is sought for data points. A search radius of 20 nm was applied, motivated by the typical maximum linkage error

Third, a Bayesian scoring as a density-overlap approach in which the experimental localization cloud is represented as a Gaussian and compared against a model density derived from the AV coordinates.

All three strategies were implemented in Python and integrated into IMP as restraint objects callable within the model-fitting framework.

# 3. Results

Kommentiert [DR13]: Allgemein halten (ohne frequentistisch vs. bayes) und dann vielleicht das kapitel3.4 allgemeiner halten: IMP + frequentistisch mit optimieren.

For this work different scoring functions were created within the IMP framework and assessed for efficiency. All scoring is based on a comparison between experimental ShareLoc data and a Computed Structure Model (CSM) from the Protein Data Bank as benchmark. The scoring functions differ in their mathematical formulation and their computational approach.

## 3.1. Simple Distance measurement

Kommentiert [DR14]: Ist eine summe an scores zwischen punkten => gut für punktschätzer.

The first scoring function compares experimental localizations pointwise with the model's Avs regarding their distance (Wu et al. 2023). This is based on a squared Euclidean distance. Each comparison is weighed by its precision. Comparisons with high uncertainty contribute less to the total score. A Euclidean distance that takes variance into account is a Mahalanobis distance. The score is defined as a negative log-likelihood derived from a Gaussian error model. For a set of experimental localizations $x_{D,i}$ and model AV positions $x_{M,j}$, the score $S$ is calculated by summing the contributions of all pairs:

$$Score = \sum_i \sum_j -0{,}5 \frac{(x_{M,j}-x_{D,i})(x_{M,j}-x_{D,i})}{(\sigma_{D,i}+\sigma_M)^2} \qquad \text{(eq. 9)}$$

where $\sigma_{D.i}$ is the localization uncertainty derived from the photon count and $\sigma_M$ the variance of the model Avs. In Python (BOX 1), the corresponding function receives a list of AVs as parameters for the model, and an array of means and an array of covariances for the experimental data.

Kommentiert [DR15]: See 2.1 PDB

```
FUNCTION ComputeScoreSimple(model_avs, data_mean, data_cov, weights =
None):
    # model_avs: list of AV particles (IMP.bff.AV), each has 3D coordinates
    # data_mean: array of N data means, each a 3D vector
    # data_cov:  array/list of N data covariance matrices, each 3x3

    score_total ← 0.0

    sigma_av ← 8.0   # fixed scalar used to build isotropic model
covariance

    FOR i FROM 0 TO length(data_mean)-1 DO
        x_D ← data_mean[i]     # 3D vector
        Sigma_D ← data_cov[i]  # 3x3 matrix
        w_D ← weights[i]

        # data-only prefactor (note: depends on det(Sigma_D) only, as in
code)
        prefactor ← w_D / sqrt( 2 * pi * det(Sigma_D) )
```

```
        FOR each av IN model_avs DO
            x_M ← Coordinates(av)  # 3D vector from
IMP.core.XYZ(av).get_coordinates()

            # isotropic model covariance
            Sigma_M ← sigma_av * IdentityMatrix3x3()

            # combined covariance used in exponent
            Sigma ← Sigma_D + Sigma_M

            # quadratic form (Mahalanobis distance w.r.t. Sigma)
            exponent ← -0.5 * Transpose(x_M - x_D) * Inverse(Sigma) * (x_M
- x_D)

            # accumulate log-score contribution
            score_total ← score_total + log(prefactor) + exponent
        END FOR
    END FOR

    RETURN score_total
END FUNCTION
```

BOX 2: TODO

This algorithm, with its two nested loops, has a complexity of $O(M * N)$ with $M$ and $N$ number of data respective model elements. Benchmarking confirms that this is rather slow. Even with small dataset sizes, the algorithm quickly takes over a second (FIG. 9). Even for small datasets the execution time easily reaches seconds.

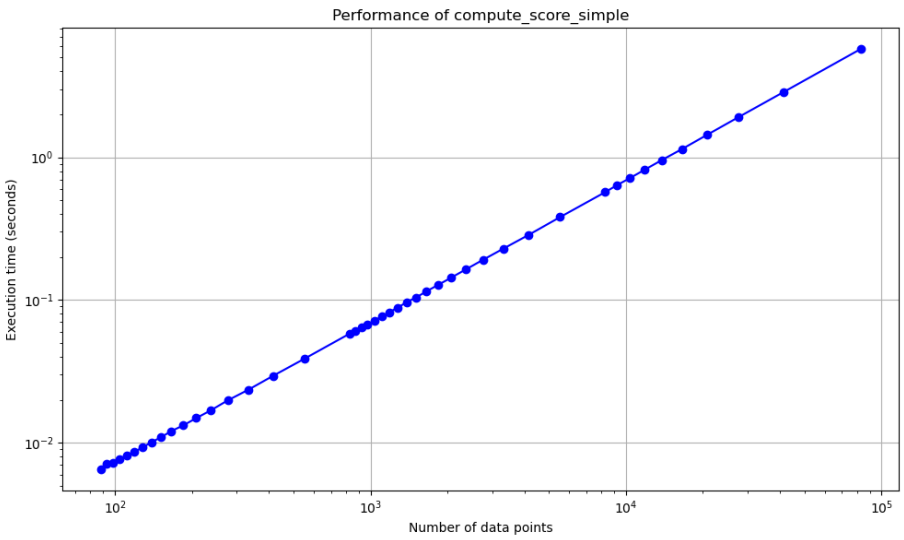



FIG. 9: TODO: - bei einem single Model NPC -> Auswirkungen auf

**Kommentiert [DR16]:** Stand jetzt: Anpassung der Model NPC an die Daten ist da nicht drin.

## 3.2. KD-Tree

To address the performance issue, a spatially optimized version was implemented and tested. In this version the used data structure was changed. Here, KD-Trees, multidimensional binary search trees, as data structure for experimental data was chosen. The data points stored in the tree-nodes can be queried by exact match, partial match or radius match. Each record in the node contains a k-tuple of attributes and pointers to the neighboring nodes (Bentley 1975). This radius search restricts the pairwise comparison to a local neighborhood. This radius r can be defined as maximum distance between molecule and fluorophores when using common dye attachment methods. The implementation was done using the KD-Tree implementation from Scikit-Learn (BOX 3).

```
FUNCTION ScoreTree(tree, model_avs, data_xyz, data_var,
                   scaling, search_radius, offset_xyz = None):

    model_xyzs = empty list
    FOR each av IN model_avs:
        xM = coordinates_of(av)
        xM = convert_to_float_array(xM)
        APPEND xM TO model_xyzs
    END FOR
    model_xyzs = as_array(model_xyzs)

    ind = tree.query_radius(model_xyzs, radius = search_radius)

    score_total = 0
    FOR avidx FROM 0 TO length(ind)-1:
        neighbor_indices = ind[avidx]
        xM = model_xyzs[avidx]

        xD = data_xyz[neighbor_indices]
        varD = data_var[neighbor_indices]

        FOR each (xDi, varDi) IN zip(xD, varD):
            di = -0.5 * dot(xM - xDi, xM - xDi) / varDi
            score_total = score_total + di
        END FOR
    END FOR

    RETURN score_total
END FUNCTION
```

BOX 3: TODO

Using a KD-Tree as a data structure improves the complexity from $O(M * N)$ when comparing all possible pairs to $O\big(M * log(N)\big)$ for KD-Trees. However, inserting the data, i.e., initializing the tree, must still be considered. The complexity here is O(log(N)). Efficiency of KD-Trees decreases with high-dimensional data which makes it more worthwhile for low-dimensional data (Bentley 1975; "KDTree" n.d.). However, there are increasingly approaches to improve efficiency even in higher dimensions (Ram and Sinha 2019; Cao et al. 2020).

Benchmarking is consistent with these assumptions. A large efficiency improvement can be achieved. Using a search radius of 20 nm, the evaluation time for the same benchmarking dataset dropped to 1.3 milliseconds which is a nearly 10,000-fold speedup (FIG. 10).

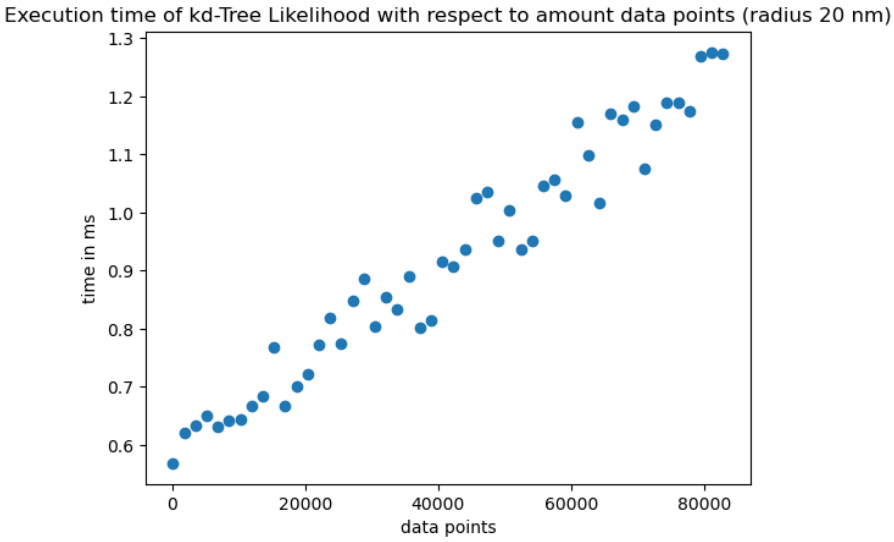



FIG. 10: TODO -> Anpassung wie oben – zusätzlich Benchmarks mit initialization of tree

Even with larger radii the execution times are stable and meke the algorithm advantageous against the simple distance scoring (FIG. 11).

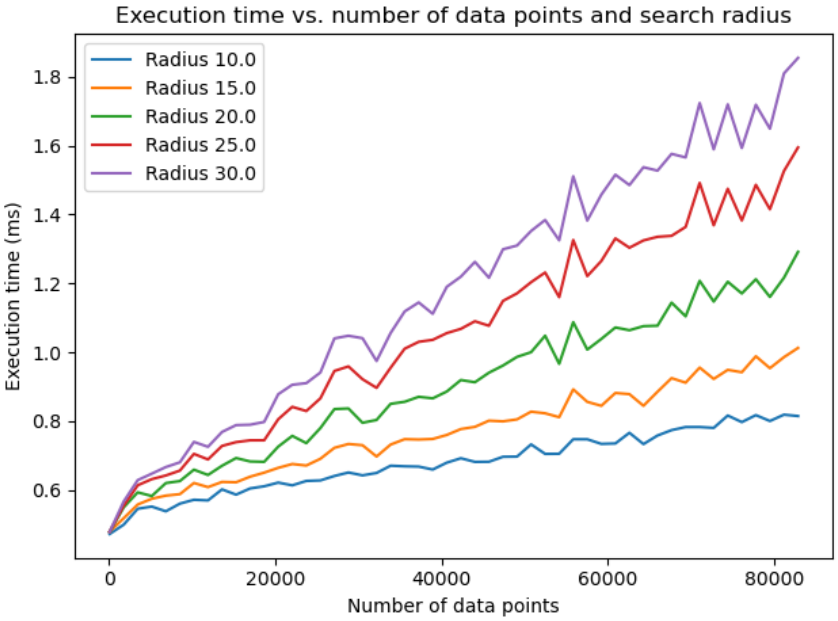



FIG. 11: TODO

This clearly demonstrates the necessity of considering the data structure when generating a likelihood function.

## 3.3. Density-based scoring

While the former scoring functions measure pointwise distances between two localizations which are summed up, another approach was integrated that uses density-based scoring. The basic idea is to use the overlap of two Gaussians instead of the distance between two data points as a measure of the model's quality (Bonomi et al. 2019) (FIG. 12).

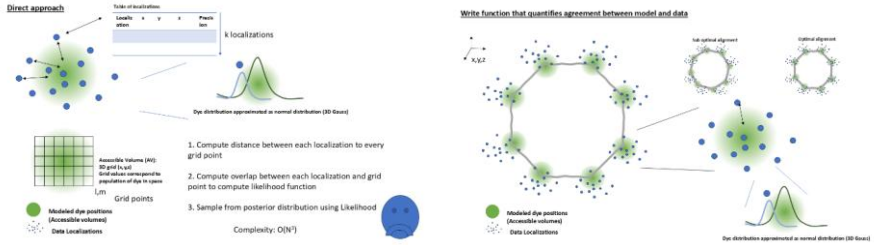



FIG. 12: TODO

The Gaussian distribution is defined as:

$$G\left(x \middle| x_{D,i}^{j}, \Sigma_{D,i}^{j}\right) = \frac{1}{(2\pi)^{\frac{3}{2}}\left|\Sigma_{D,i}^{j}\right|^{\frac{1}{2}}} exp\left[-\frac{1}{2}(x - x_{D,i}^{j})^{T}(\Sigma_{D,i}^{j})^{-1}(x - x_{D,i}^{j})\right] \qquad \text{(eq. 10)}$$

with $x$ as the localization, $x_{D,i}^{j}$ as the mean and the covariance matrix $\Sigma_{D,i}^{j}$.

The question arises whether a single distribution over all data is suitable, especially for more complex models. Gaussian Mixture Models (GMM) provide an alternative to a single distribution for all data points, as these models can form clusters that each have their own distribution for each cluster (Kawabata 2008). GMMs are a linear combination of multiple Gaussians. Each individual Gaussian, representing a component, has its own mean and covariance. All Gaussians are weighted by the mixing coefficient that determines how much this component contributes to the sum (Bishop 2006):

$$\Phi_{D}^{j}(x) = \sum_{i=1}^{j} \Phi_{D,i}^{j}(x) = \sum_{i=1}^{j} \omega_{D,i}^{j} * G\left(x \middle| x_{D,i}^{j}, \Sigma_{D,i}^{j}\right) \qquad \text{(eq. 11)}$$

where $\omega_{D,i}^{j}$ is the normalized weight (Bonomi et al. 2019).

Kommentiert [DR17]: CITATION NEEDED: alte Version anschauen.

Kommentiert [DR18]: Alternative to what

An advantage of GMM is that using components means there are no correlations between them which increases the model quality because each component is fitted independently with respect to the clusters. Furthermore, GMMs can represent all types of distributions with a relatively small number of parameters. Overlaps when comparing two Gaussian distributions can be calculated more quickly since mathematical procedures such as gradient-based optimizations can be applied (Kawabata 2008; Bonomi et al. 2019). Since there are not only the Gaussians' parameters that must be determined but also latent variables for the weight of each component, a solution can only be found iteratively. An algorithm that enables this is the Expectation-Maximization algorithm (EM-algorithm). The name was first used by (Dempster, Laird, and Rubin 1977); however, its origins can be traced back to the 1920s (Fisher 1925; Meng and Van Dyk 1997). Since the 1980s, the EM algorithm has been used for parameter estimation in GMMs (Kawabata 2008). The algorithm is iterative. Initially, random parameters are used to form clusters during the expectation step by a clustering method that is like k-means clustering. Then the parameters are optimized for each cluster during the maximization step. A potential issue with the EM algorithm is slow convergence to a maximum depending on the choice of initial parameters. However, there are avoidance strategies for this (Meng and Van Dyk 1997). Additionally, it is not guaranteed that the global maximum will be found rather than just a local maximum (Bishop 2006). Another issue of this algorithm is that the number of components must be determined in advance. In general, the higher the resolutions or the higher the complexity, the more components are required to describe the structure best (Kawabata 2008). There are several options for choosing the optimal number beforehand. A common method is the application of information criteria for that purpose (McLachlan and Rathnayake 2014; "2.1. Gaussian Mixture Models" n.d.). Two information criteria that are based on MLE are the Akaike Information Criteria (AIC) and the Bayesian Information Criteria (BIC). Both penalize the number of parameters of the model to avoid overfitting but BIC favors simpler models more. BIC is defined as

$$BIC = -2\ln\big(L(\theta|x)\big) + \delta \log n \qquad \text{(eq. 12)}$$

where $\ln\big(L(\theta|x)\big)$ is the logarithm of the likelihood function, $n$ is sample size and $\delta$ is the number of free parameters in the model (MacKenzie et al. 2018). The python library Scikit-Learn offers a handy implementation for GMM. When fitting a model this implementation directly gives the average log-likelihood and scores for information criteria for the applied number of components. These scores can be used to determine an optimal number (BOX 4).

```
function SELECT_GMM_COMPONENTS(data, component_min, component_max):

    # data: array of shape (N, D), e.g. D=2 or D=3 coordinate point cloud

    # Build candidate list (log-spaced / powers of 2, as in your code)
    K_list = []
    K = max(1, component_min)
    while K <= component_max:
        append K to K_list
        K = K * 2

        for r in 1..n_restarts:
```

```
            gmm = GaussianMixture(n_components=K, covariance_type=cov_type)
            gmm.fit(data)

            score = gmm.score(data)  # average log-likelihood per sample
            aic   = gmm.aic(data)
            bic   = gmm.bic(data)

            # Keep the best run for this K (criterion can be "lowest BIC")
            if bic < best_bic_for_K:
                best_bic_for_K   = bic
                best_aic_for_K   = aic
                best_score_for_K = score
                best_model_for_K = gmm

        append (K, best_bic_for_K, best_aic_for_K, best_score_for_K,
best_model_for_K) to results

    # 3) Select K* via minimum BIC (as in your code)
    (K_star, _, _, _, gmm_star) = argmin_over_results(results, key =
best_bic_for_K)

    # 4) Return K* and fitted parameters needed by the scoring
    means        = gmm_star.means
    covariances  = gmm_star.covariances
    weights      = gmm_star.weights

    return { "K_star": K_star,
             "results_per_K": results,
             "gmm_star": gmm_star,
             "means": means,
             "covariances": covariances,
             "weights": weights }
```

BOX 4: TODO Pseudocode

A comparison between average log-likelihood, AIC and BIC shows the stronger penalizing of free parameters for BIC in comparison to the other measures (FIG. 13).

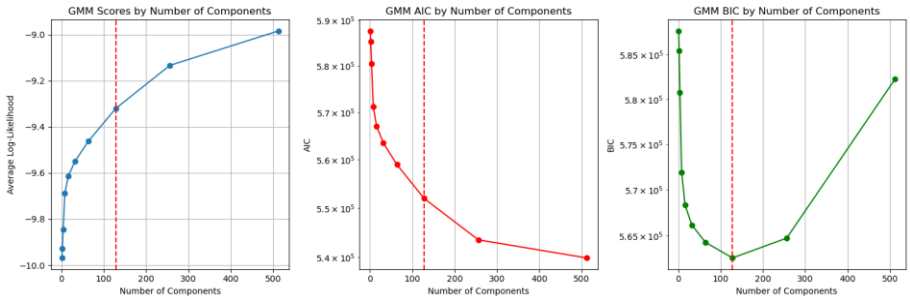



FIG. 13: TODO – overfitting nur BIC und AIC

The processing of PDB data through IMP has already been discussed regarding the system representation in the methods chapter. The AVs created by IMP can be viewed as Gaussian distributions, as the exact location of the fluorophore is uncertain and can only be said to be

within the AV. From this, the mean and sigma can be calculated, which are used as parameters in the Gaussian.

As forward model the coordinates of the accessible volumes of the PDB data are modelled as a density of several Gaussians.

```
model_xyzs = np.array([IMP.core.XYZ(av.get_particle()).get_coordinates()
for av in self.avs_decorators])
```

BOX 5: TODO – Forward model

The noise model, which reflects the deviation between data and model, is defined as:

$$\int dx\, \Phi_{M,j}(\mathrm{x})\Phi_{D,k}(\mathrm{x})$$

$$= \frac{\omega_{M,j}\omega_{D,k}}{(2\pi)^{\frac{3}{2}}\left|\Sigma_{M,j}+\Sigma_{D,k}\right|}\exp\left[-\frac{1}{2}(x_{M,j}-x_{D,k})^T(\Sigma_{M,j}+\Sigma_{D,k})^{-1}(x_{M,j}-x_{D,k})\right] \quad (18)$$

Sigma can be written as the sum of the model and data covariance, as it involves two GMMs (Bonomi et al. 2019). For simplification, the logarithm is again used for calculation. In Python the GMM scoring is implemented as a sum over all data points (BOX 6).

**Kommentiert [DR19]:** Komplett überarbeiten nach neustem code:

```
def compute_nb_gmm(
        model_xyzs,
        data_mean,
        data_cov,
        data_weight
):
    print("gmm_score.py - compute_nb_gmm")
    score_total = 0.0

    # Model loop
    for model_xyz in model_xyzs:

        # placeholder for AV covariance
        sigma_av = 8.0
        sigma_M = np.eye(3, dtype=np.float64) * sigma_av

        # Data loop
        for i in range(len(data_mean)):
            print("Start GMM Restraint - Loop: ", i)
            mean_D, sigma_D, weight_D = data_mean[i], data_cov[i],
data_weight[i]
            Sigma_D = sigma_D
            Sigma_M = sigma_M
            weight_M = 1.0
            xD = mean_D
            xM = model_xyz
            Sigma = Sigma_D + Sigma_M  # Bonomi et al. 2019, Bayesian EM
eq. 12
            prefactor = (weight_M * weight_D) / ((2. * np.pi) ** (3. /
2.) * np.linalg.det(Sigma) ** 0.5)
            exponent = -0.5 * (xM - xD).T @ np.linalg.inv(Sigma) @ (xM -
xD)
            score_total += np.log(prefactor) + exponent
            print(score_total)
```

```
    return score_total
```

BOX 6: TODO

Since matrices multiplications with Python are slow because of their complexity, the just-in-time compiler from the Python Numba library was used to enhance efficiency and runtime. Here, the code is compiled to faster C code for faster matrices multiplication ("Compiling Python Code with @jit — Numba 0.52.0.deV0+274.G626b40e-Py3.7-Linux-X86_64.Egg Documentation" n.d.).

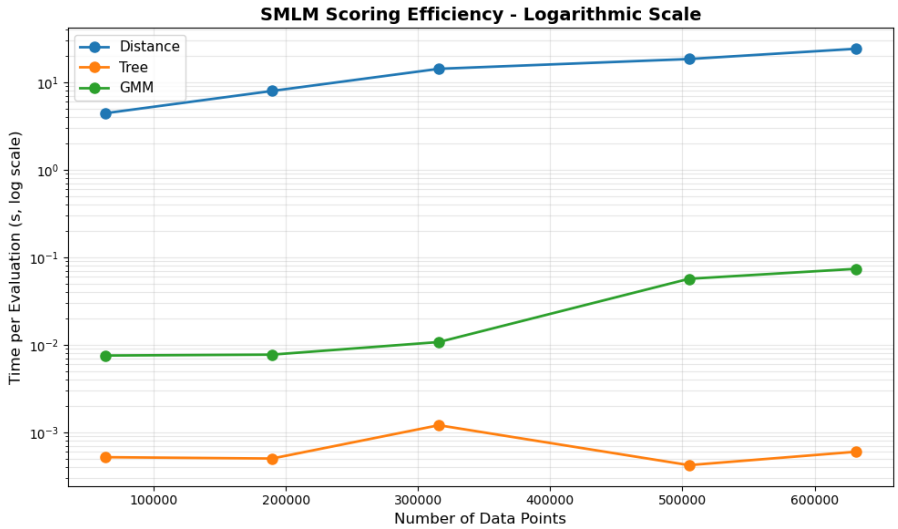



FIG. 14: benchmark_plotter.py – including initialition time – check with benchmark above

While the GMM scoring evaluation is significantly faster than the naive distance score, it is computationally more intensive than the KD-Tree approach due to the matrix algebra involved in form of calculating determinants and inverses (FIG. 14). Additionally, the initialization phase, calculating the optimal number of components and fitting the model to these components, takes additional time since the EM-algorithm must be applied to all number of components that are fitted. The complexity of GMM is $O(NKD^2)$ with $N$ data points, $K$ components and $D$ dimensions (Pinto and Engel 2015). This aligns with the measurements.

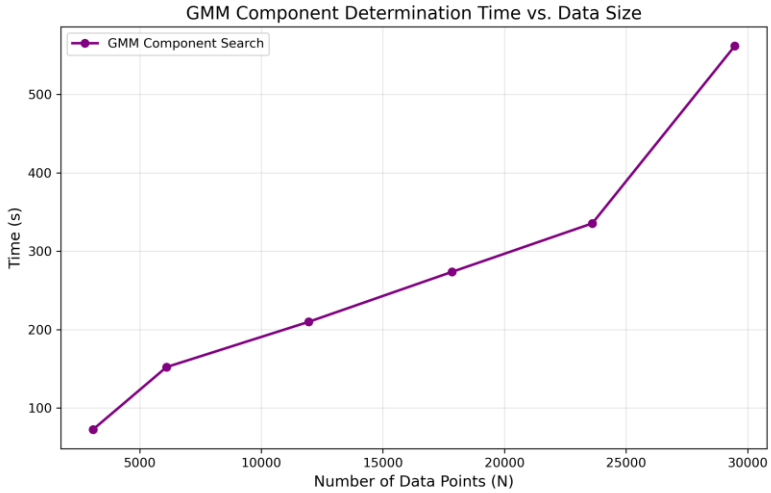



FIG. 15: Time for initializing subject to data size

## 3.4. Model and restraints in IMP

Before, the IMP library was already used for generating accessible volumes as part of the system representation. There, an accessible volume for the model was defined that was generated with prior knowledge of the physical properties of dyes which restrain the space where these dyes can be attached, the solution space.

Here, the scoring functions are encapsulated as IMP-restraints into the model. A wrapper class allows the user to choose the scoring function that should be used (BOX 7). Besides, it is also possible to use a restraint set that can be assembled as a weighted sum for several restraints included. For example, further restraints can be based on distance, density or excluded volumes. Each term can be weighed in respect to noise of the particular data. In this case a spatial restraint for scoring was implemented that measures the fit of experimental data to the model data (Saltzberg et al. 2021; Sali 2021; Russel et al. 2012; Bonomi et al. 2019).

```
class ScoringRestraintWrapper(IMP.pmi.restraints.RestraintBase):
    def __init__(self,
                 m: IMP.Model,
                 avs: typing.List[IMP.bff.AV],
                 gmm_sel_components=None, gmm_sel_mean=None,
gmm_sel_cov=None, gmm_sel_weight=None,
                 dataxyz=None, var=None, kdtree_obj=None,
                 weights=None,
                 scaling: float = 1.0, searchradius: float = 10.0,
offsetxyz: np.ndarray = None, # Adjusted default scaling
                 type: str = "GMM",
                 label: str = "Scoring Restraint"):
```

```
        self.rs = IMP.RestraintSet(m, 'ScoringRestraint')
        self.type = type
        self.scoring_restraint_instance = None
```

```
class ScoringRestraintGMM(IMP.Restraint):
    def __init__(self, m, avs: typing.List[IMP.bff.AV], gmm_sel_components,
gmm_sel_mean, gmm_sel_cov, gmm_sel_weight):


[…]

def unprotected_evaluate(self, da):
    print("scoring_restraint.py - ScoringRestraintGMM -
unprotected_evaluate")
    model_xyzs =
np.array([IMP.core.XYZ(av.get_particle()).get_coordinates() for av in
self.avs_decorators])
    score = compute_nb_gmm(model_xyzs, self.gmm_sel_mean, self.gmm_sel_cov,
self.gmm_sel_weight)
    return score
```

BOX 7: TODO

Gathering information and generating a system representation and scoring function, which are shown here, are the first steps of a typical IMP modeling process. In the subsequent steps, the posterior for the representation can be calculated, e.g. with sampling methods like Monte Carlo Markov Chain (MCMC) or the conformational space can be explored with simulation methods like Brownian Dynamics which can be used for stochastic movement of particles in liquids as it happens with molecules in the cell's cytoplasm (Huber and McCammon 2019; Schlick 2010; Sali 2021; "IMP Tutorial: Brownian Dynamics in IMP" n.d.; "IMP::Core::MonteCarlo Class Reference" n.d.; Saltzberg et al. 2021; Früh et al. 2021). This is the third step for which IMP offers a range of methods in its library (Russel et al. 2012; "IMP Tutorial: Brownian Dynamics in IMP" n.d.). In the last step, the resulting ensemble of good-scoring models is analyzed to determine the precision of the final structure and to validate it against data that was not used in the modeling process (Saltzberg et al. 2021; Russel et al. 2012).

**Kommentiert [DR20]:** Evtl. den abschnitt in 4

**Kommentiert [DR21R20]:** Achtung vielleichtden abschnitt davor auch.

# 4. Discussion

In the previous chapter, three scoring functions were developed. The functions were also discussed regarding their mathematical formulation and their computational efficiency.

The simple distance scoring and the KD-Tree scoring, as a special case with changed data structure for the data, as well as GMM scoring, both measure the consistency between SMLM data and the NPC structural model. However, they differ in the ways they account locally and globally for the data, they incorporate uncertainty and their suitability for frequentists or Bayesian approaches.

In the following chapter these scoring functions are discussed considering the theoretical concepts described in Chapter 1. There are several key questions arising from the previous work. What are the roles that distance and KD-Tree scoring can play in frequentist tools? Which additional steps must be integrated into the GMM modeling workflow for a Bayesian posterior formulation as proposed by Bonomi/Latham? How well do these approaches capture experimental uncertainties and systematic errors? Which limitations arise from the observed runtime and scaling behaviors?

## 4.1. Comparison of models

As most direct implementation the simple distance scoring was introduced as first approach. It is a Gaussian-based likelihood between model and data localizations. Mathematically, each term corresponds to a negative log-Gaussian factor, where the squared distance between a model point (AV center) and an experimental localization is scaled by the sum of their variances. This scoring function is closely related to the likelihood approach of Wu et al., which relies on a product-form Gaussian likelihood of individual localizations.

The uncertainty in this approach is directly integrated into the score via variance $\sigma$ which is derived from the amplitude of the localizations' photon count for the data and the AV for the model. The weighted sum of Gaussian log-likelihood with both variances as sum in this mathematical formation makes it well suited for optimization and MLE in a frequentist context. Therefore, this fitting of single structures is best for finding a single best model. It is less suitable for approaches that need less rigid, more flexible ensembles like dynamic systems that have often no one-to-one mapping of points.

This exhaustive testing of all data points with all model points makes this approach with its computational complexity of $O(M * N)$ not useful for realistic datasets. This was underlined with the runtime benchmark of this algorithm.

The KD-Tree scoring adopts the same pointwise Gaussian likelihood as the distance scoring but replaces the exhaustive pairing of all data and model points with a local neighborhood search using a KD-Tree as data structure. Instead of comparing every AV position with all SMLM localizations, only data points within a predefined search radius are considered. Mathematically, the score remains a negative log-likelihood term over independent Gaussian contributions. The measured runtimes demonstrate that this change in the data structure is a crucial step to decrease the computational time for large datasets massively without requiring a change to the underlying likelihood form. The computational complexity decreases from $O(M * N)$ to $O\big(DNlog(N)\big)$. This makes the KD-Tree scoring a realistically applicable scoring approach for optimization. This could be done within the IMP context, for instance, for the rapid initialization of models. At the same time, the fundamental limitation remains that uncertainty is modeled only pointwise and global density properties are not captured.

The GMM Score differs fundamentally in concept from the Distance and KD-Tree approaches. Instead of evaluating individual point distances, it compares the overlap of two continuous densities. The ShareLoc localization data is decomposed into clusters, the components, for

which individual means, covariance matrices, and weights are estimated. Analogously, the Accessible Volumes of the NPC model are converted into a GMM representation. The score then calculates analytical overlap terms similar to the Bayesian map-likelihood in Bonomi et al.

Uncertainty is integrated for each overlap as covariance matrix of the sum of data and model covariances. Each GMM component has its own covariance matrix. This shift is particularly valuable for IMP, as it often employs "coarse-grained" models, simplified representations where a protein is not represented atomically but as a volume or sphere. A GMM is a natural mathematical equivalent for such fuzzy models.

As expected in the theoretical considerations, the algorithm is fast in this case. Here, with just three-dimensional data this is not a big aspect in the algorithm's computational complexity of $O(NKD^2)$ with $N$ data points, $K$ components and $D$ dimensions. Nonetheless, the benchmark showed that the algorithm is still slower than the limitation of possible comparisons via KD-Tree.

Since GMM fitting needs the number of different components beforehand, this number must be calculated in the algorithm's initiation process. Here, the Bayesian Information Criteria was used. This extensive initiation phase prolongs the algorithm's runtime.

## 4.2. Frequentist and Bayesian Utilization of Scores

The three scoring functions support different statistical roles. The Distance and KD-Tree scores implement a pointwise likelihood $p(D|X)$ suitable for frequentist point estimators (MLE/MAP). Here, a single best-fit model is sought.

The GMM score, in contrast, is inherently designed for Bayesian usage. It provides a likelihood term that can be seamlessly combined with priors. In a complete Bayesian pipeline, the scoring function would not be used to determine a single model, but rather to generate an ensemble of structures through sampling. This Bayesian posterior $p(X|D) \propto p(D|X)\, p(X)$ directly represents uncertainty as a distribution. An approach that offers advantages over purely frequentist point estimators for underdetermined systems.

## 4.3. Handling Uncertainty and Validation Perspectives

Regarding the separation of precision (variance) and bias, the distance and KD-Tree scores primarily address statistical uncertainty via localization precision derived from the photon intensity in the data.

The GMM score offers more flexibility because covariance matrices also encode local density inhomogeneity and potential correlations between clusters. For complete validation, a simulation benchmark would be required, evaluating synthetic SMLM data with known ground truth. It is expected that the GMM score would be more robust with highly noisy data compared

to the hard Distance Score, as it compares "softer" density distributions and is less sensitive to individual outlier points.

## 4.4. Limitations of the Work and Outlook

This work intentionally focuses on the implementation of the scoring functions, not on a complete Bayesian inference chain. Neither posterior distributions over model parameters are explicitly calculated, nor are ensemble analyses performed.

Some simplifications were used which need further work. First, the scoring functions tested in this work are applied to a global set of localizations that contains many copies of the NPC, while the structural model represents only a single, idealized NPC. A realistic application would require e.g. a preceding identification and registration of individual NPC particles to fit the model to each particle separately as done by Wu et al. Second, in the current implementation, the model uncertainty of the accessible volumes is approximated only by a fixed, heuristic covariance $\Sigma_M$ that is not explicitly derived from the AV geometry. A more rigorous treatment would require determining $\Sigma_M$ from the actual AV distribution, e.g. from the spread of the resampled AV points, or from the Avs' parameters like linker length or radius.

The complex matrices operations showed that it is necessary to use low-level code to improve the GMM's runtime significantly. Another option to both improves computational efficiency and approximation quality, which has not yet been implemented here, is a spatial decomposition or 'divide and conquer' strategy as in the work of Bonomi et al. The current GMM implementation uses a single global fit for the entire density. The separation of the density into subregions with separate GMMs can improve both.

Clear next steps arise from these limitations. In the short term, the heuristic parameters can be changed for valid parameters and systematic validation with synthetic data should quantitatively evaluate bias and precision of the models. In the medium term, the developed GMM score could be integrated as a likelihood term into a full IMP-Bayes pipeline to incorporate SMLM data on equal footing into integrative models.

## 4.5. Choice of Preferred Scoring Approach

In summary, the Distance and KD-Tree scores are suitable as efficient, frequentist tools for MLE estimations and initialization steps. The GMM score, however, offers the best compatibility with fully Bayesian modeling. It operates on densities, integrates uncertainty via covariances, and fits ideally with the fuzzy representation of coarse-grained models in IMP. For the long-term goal of embedding SMLM data into dynamic NPC models, the GMM-based approach is therefore the preferred choice but requires more work because of the complexity and the initialization process.

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